Fourier Analysis

Physical Intuition

Radar Detection, Sound Processing

Conducted By

Partho Sutra Dhor

Faculty, Mathematics and Natural Sciences

BRAC University, Dhaka, Bangladesh

Fourier Series

What is a Series of functions

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Fourier Series

If $f(x)$ is a continuous-time periodic signal, then the Fourier series is a series of the form

$$f(x) = \underline{constant} + \underline{(Cosine\ Terms)} + \underline{(Sine\ Terms)}$$

Fourier Series Example

$$f(x) = |x| \qquad -\pi < x < \pi$$

$$\mathcal{F}(x) = \frac{\pi}{2} + \frac{-4}{\pi}\cos(x) + \frac{-4}{9\pi}\cos(3x) + \frac{-4}{25\pi}\cos(5x) + \cdots$$

Mathematics of Fourier Series

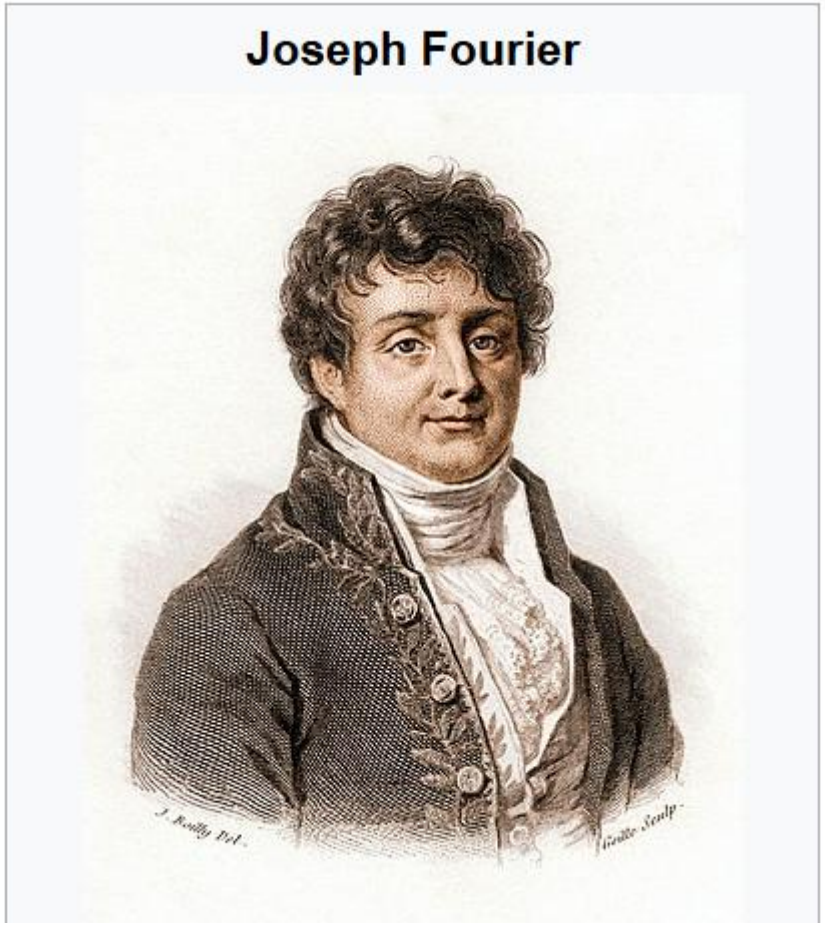
If $f(x)$ is a continuous-time periodic signal with period 2L, then the Fourier series is a series of the form

$$f(x) = \frac{a_0}{2} + \left(\sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L}\right) + \left(\sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{L}\right)$$

$$a_n = \frac{1}{T}\int_{-T}^{T} f(x)\cos\left(\frac{n\pi x}{L}\right)dt \qquad b_n = \frac{1}{T}\int_{-T}^{T} f(x)\sin\left(\frac{n\pi x}{L}\right)dt$$

Historical Background

heat Equation



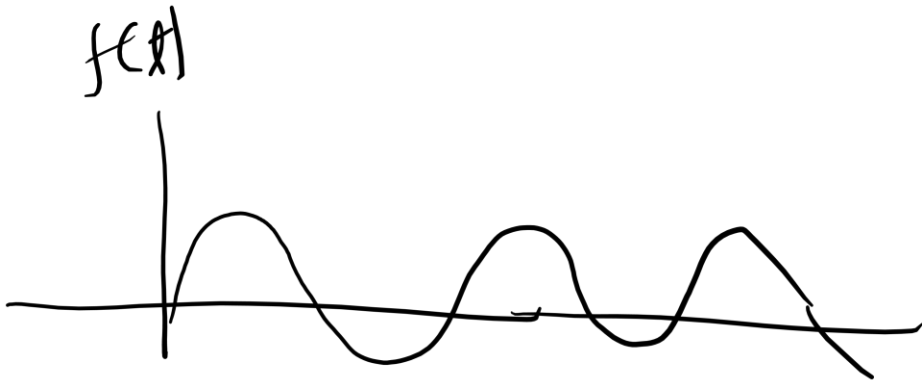
RADAR Mechanism Using Fourier Analysis

What is a Signal

→ function

↓

→ signal $f(t)$



Wave

$$f(t) = A \sin(\omega t + \phi)$$

ϕ → phase

ω → $2\pi f$ → frequency

$$A \sin(2\pi f t + \phi)$$

What is Fourier Analysis

110 Hz

$$\boxed{\sin(2x)}$$

$$\boxed{\sin(2x) + \underbrace{\sin(5x) + \sin(7x)}_{\text{Noise}}}$$

$$\int_{-\pi}^{\pi} \left[\sin(2x) + \sin(5x) + \sin(7x) \right] \cdot \sin 2x \, dx$$

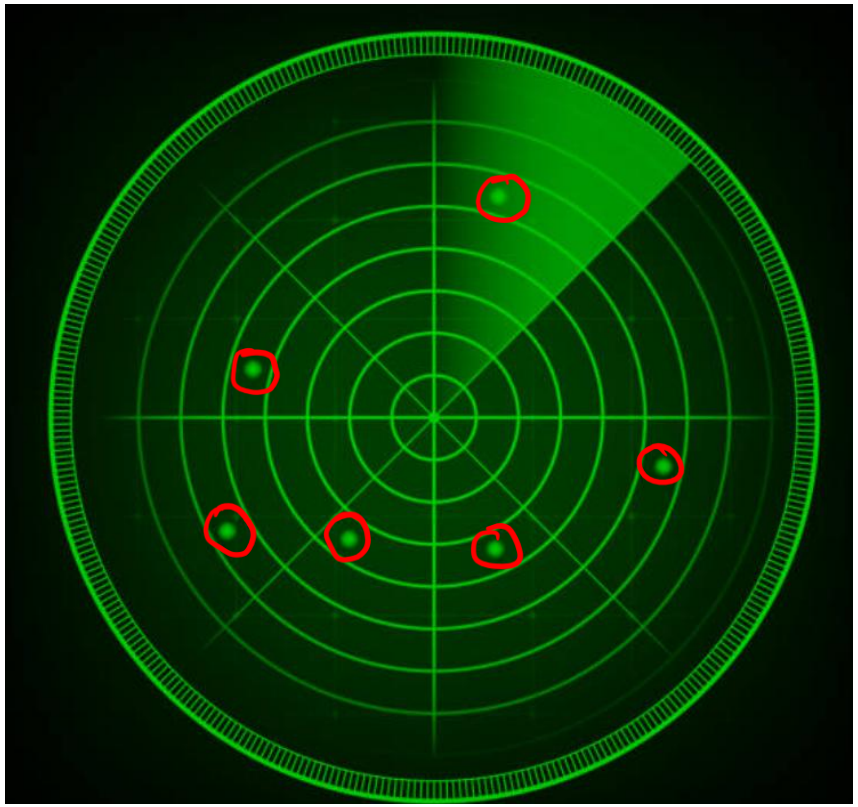
$$= \int_{-\pi}^{\pi} \sin 2x \cdot \sin 2x \, dx + \int_{-\pi}^{\pi} \sin 5x \cdot \sin 2x \, dx + \int_{-\pi}^{\pi} \sin 7x \cdot \sin 2x \, dx$$

$$= \pi + 0 + 0 = \pi \neq 0$$

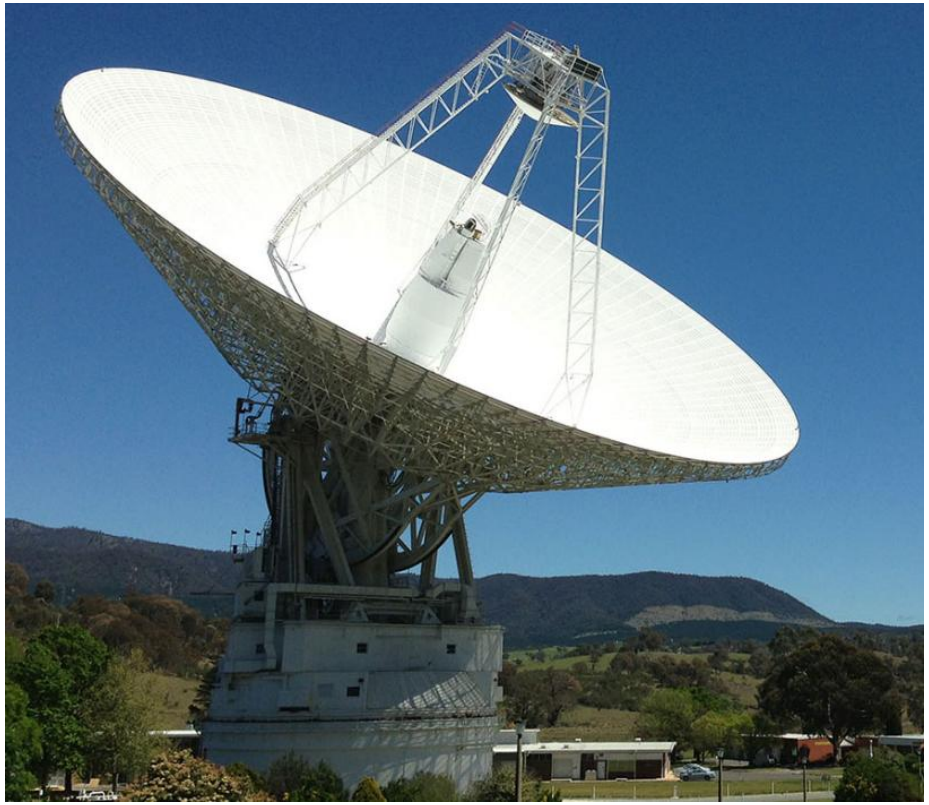
$$\int_{-\pi}^{\pi} \left[\underline{\sin 5x} + \underline{\sin 7x} + \underline{\sin 9x} \right] \cdot \underline{\sin 2x} \, dx$$

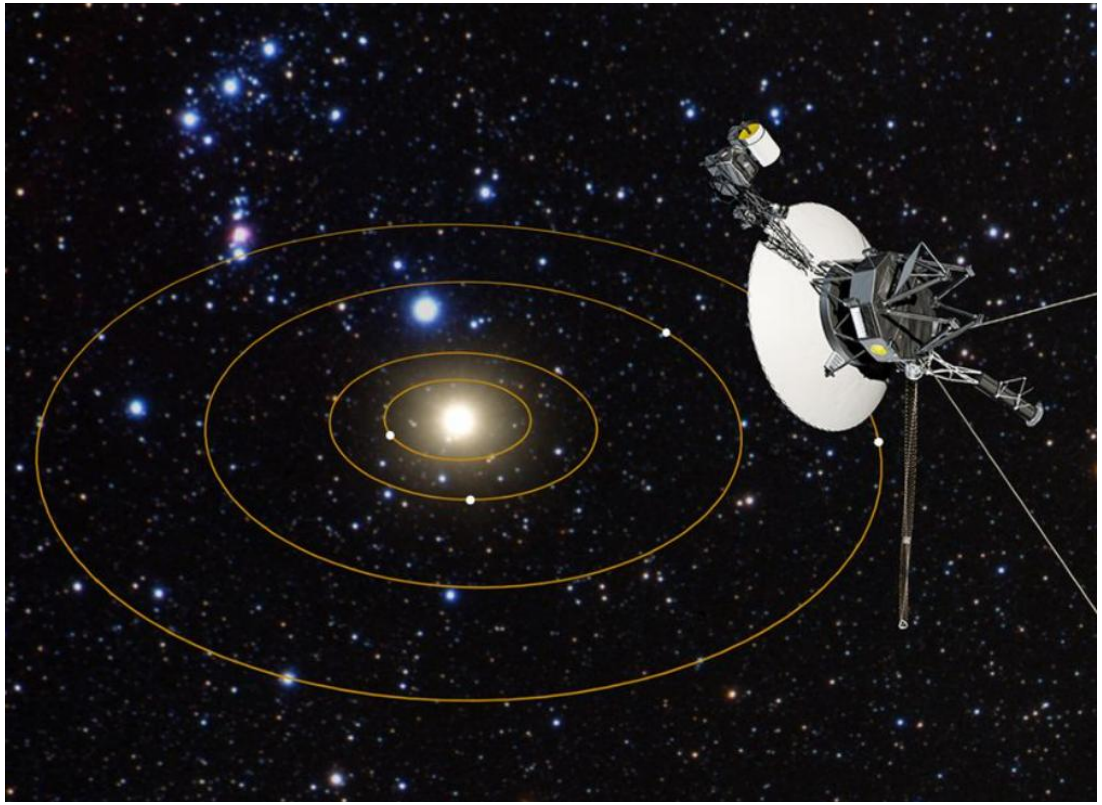
$$= 0 + 0 + 0 = 0$$

Mechanism of RADAR

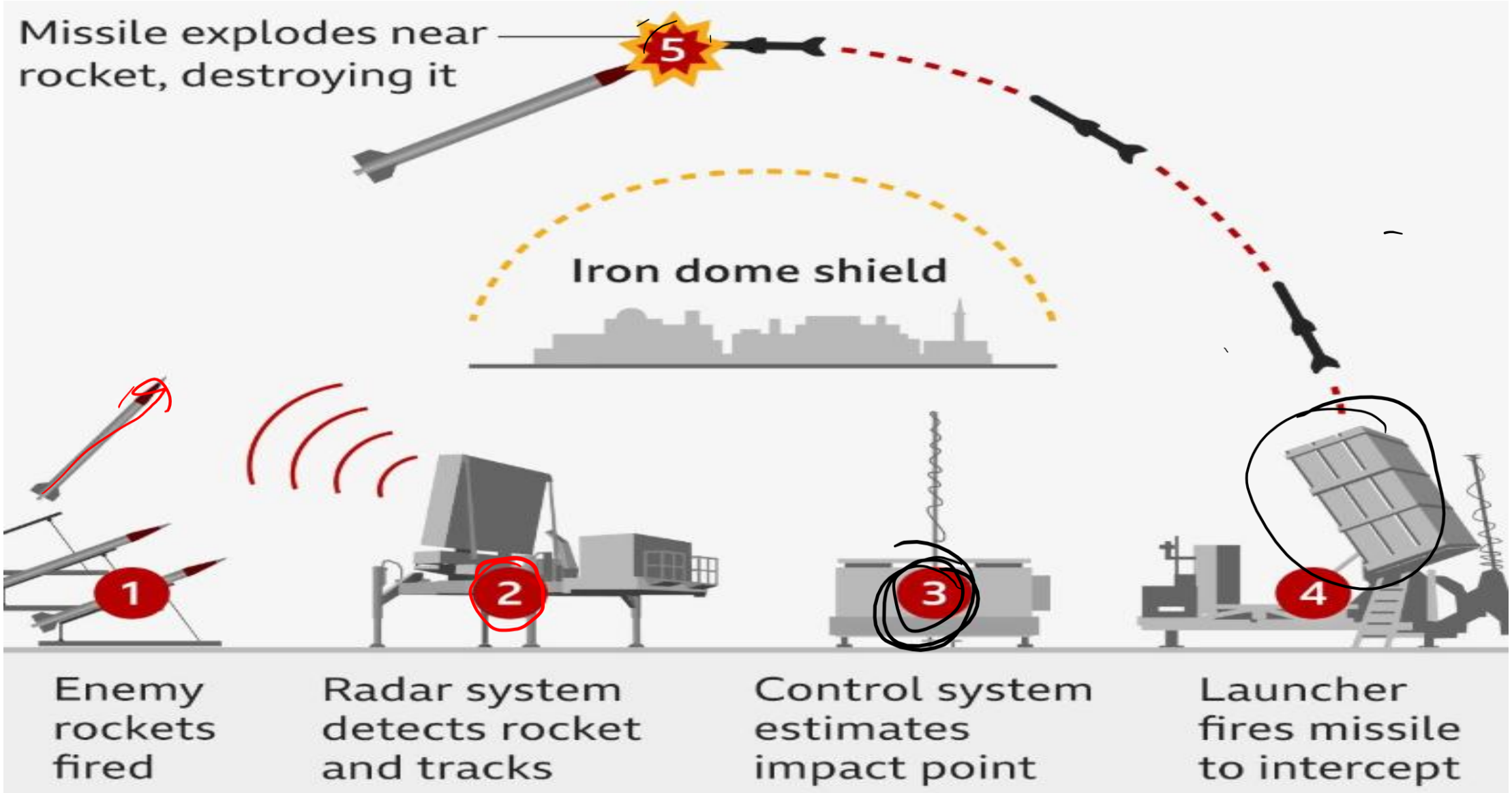


Deep Space Network





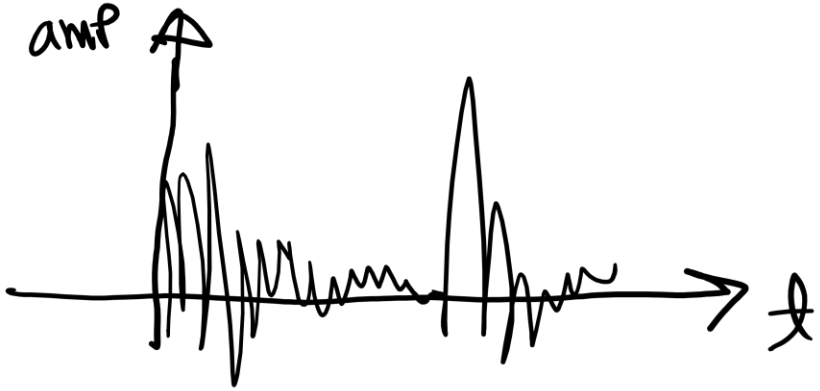
Air Defense System



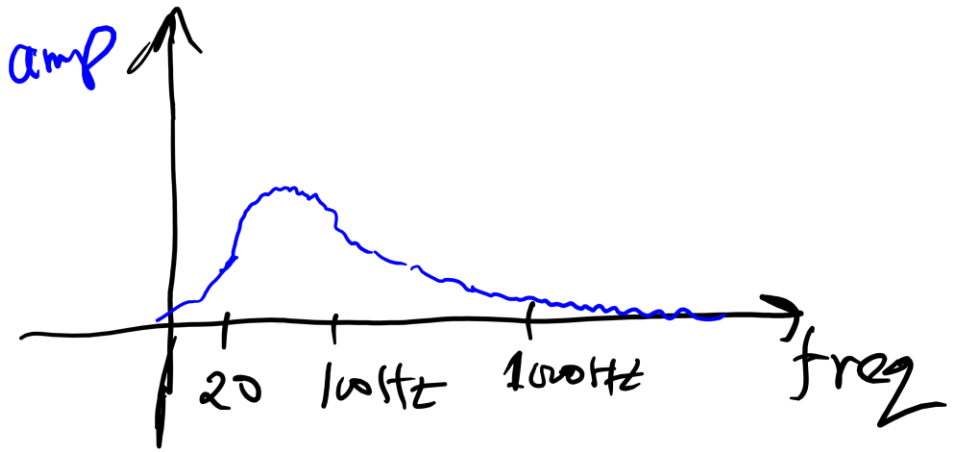
Fourier Transform

Time Domain vs Frequency Domain

$$\underbrace{f(t)}_{\text{time domain}} \xrightarrow{F.T} \underbrace{F(\omega)}_{\text{frequency domain}}$$



$\Longrightarrow$



Fourier Transform

$$\underbrace{f(t)}_{\text{time domain}} \Longrightarrow \underbrace{F(\omega)}_{\text{freq domain}}$$

Mathematics of Fourier Transform

Fourier Transform is the continuous setting of Fourier Series. Fourier Transform is a transformation technique which transforms signals from continuous time domain to the corresponding frequency domain and vice-versa. Mathematically, the Fourier transform of a continuous-time signal $f(t)$ is defined by

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} dt$$

Sound Processing
Using Fourier Transform

What is Noise Cancellation

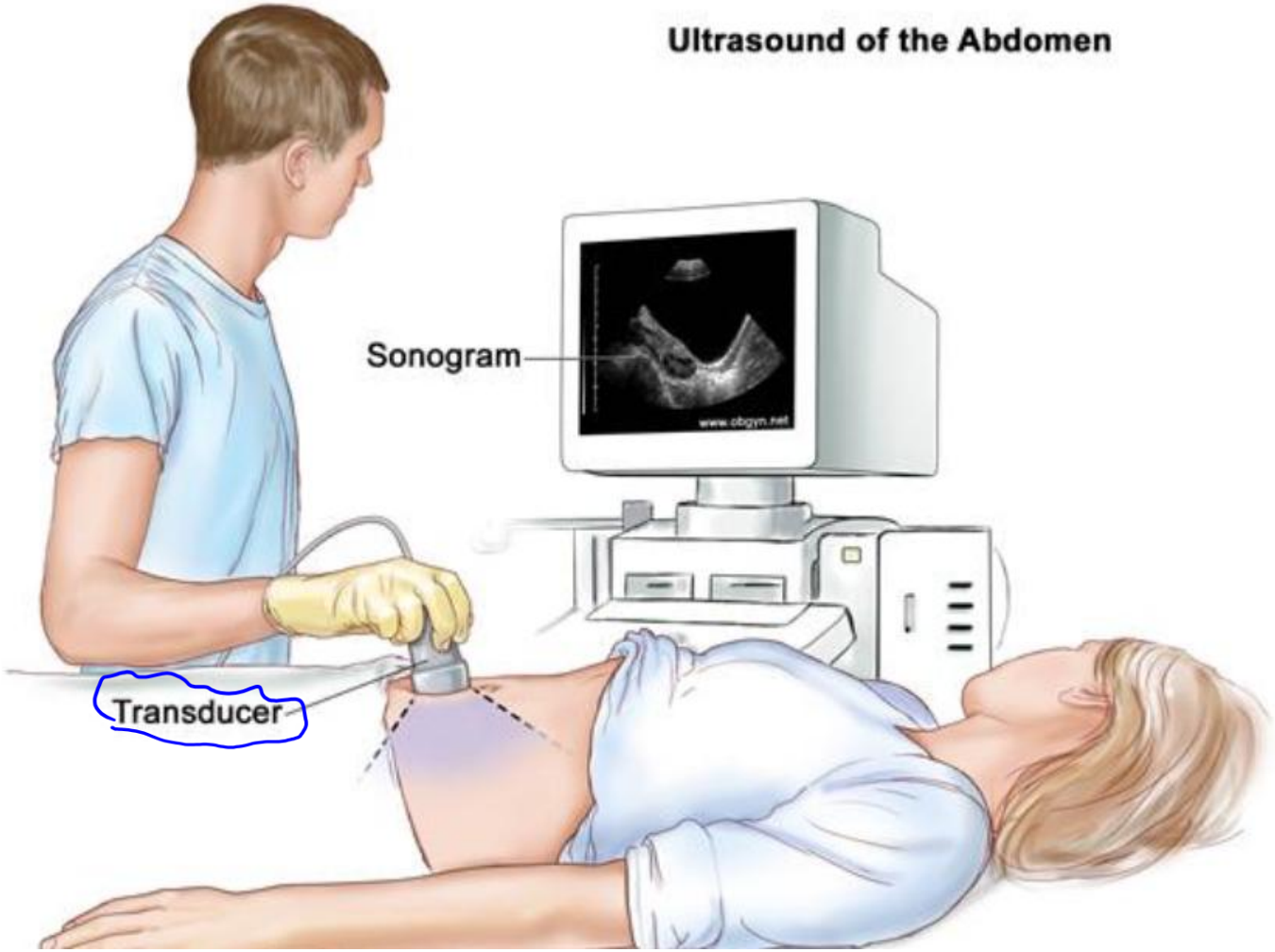
Sound + ~~noise~~ ⟹ Sound

What is Active Noise Cancellation (ANC)

TWS

Image Processing

Ultrasonography



MRI

MELBOURNE BRAIN CENTRE...
STUDY 1
Manipulated, Fused, 3D
PHR
RAH

Thank you!